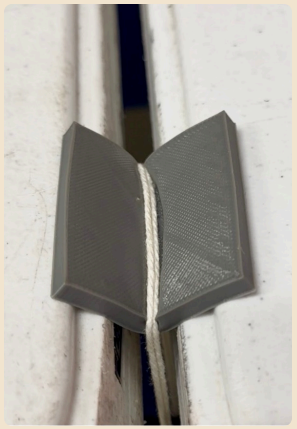
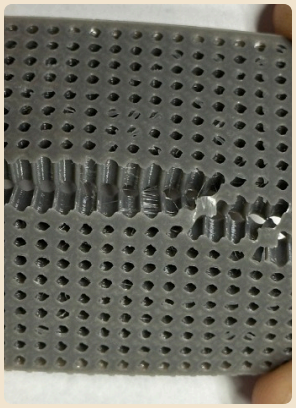
FUNCTIONS

HOW IT WORKS

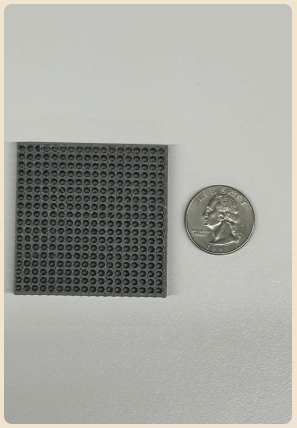
1. **Scan**
   Scan the injured astronaut's lower leg.
2. **Print**
   Print the custom shell and straps on the habitat printer.
3. **Fit**
   Unlatch the splint and fit it on vertically.
4. **Secure**
   Wrap the straps in an X, three times around the ankle.
5. **Adjust & clean**
   Loosen or tighten one-handed; remove for hygiene.



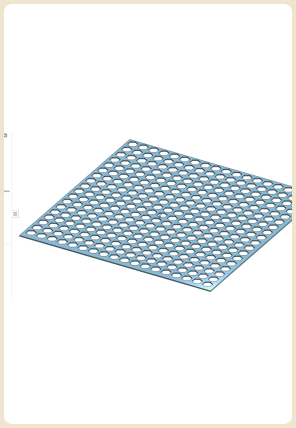
Hanging-weight test



Printed sample, split along the holes



TPU sample, quarter for scale



CAD test sample

MEET THE TEAM

TEAM LUNOVA

Gertrudis Hernandez
Prototype tester, design coordinator, expert outreach, brainstorm sketches

Jose P. Hernandez
CAD, technical drawings, 3D printing specialist

Nancy De La Cruz
Prototype tester, managing editor, sketches and designs

SURVEY OF 56 STUDENTS

Ankle injured most	63%
Can't find a good fit	54%
On their feet a lot	96%
Devices help healing	98%

WHAT THE MOON CHANGES

No hospital nearby
A return to Earth takes days, so the splint must last the whole healing time.

Every gram costs about $1,000
Stockpiling splints in every size is not practical.

Weeks of wear
Skin contact, hygiene and one-handed use matter in a closed habitat.

Scan for all our digital records

Engineering notebook (D.A.A.N.), problem statement, survey results, CAD, testing and presentation

gertrudish978-pixel.github.io/lunar-splint

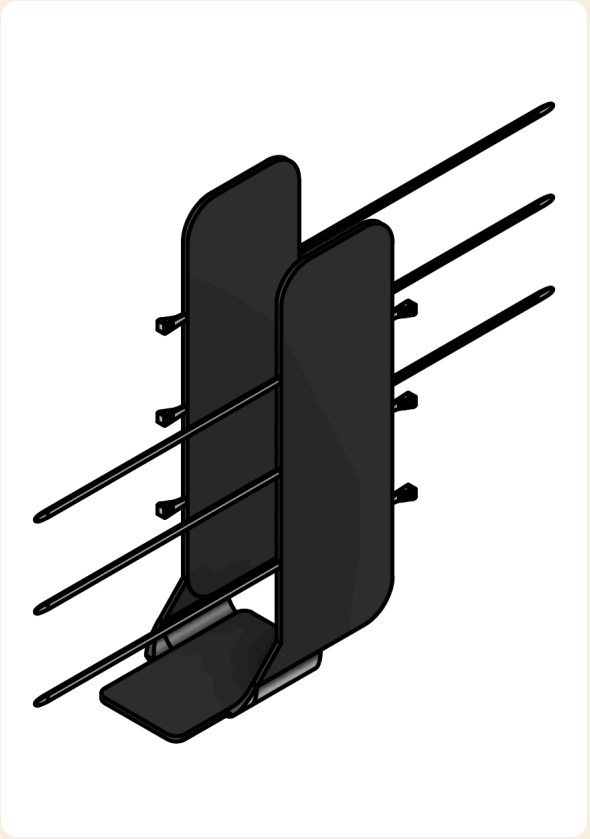
A NASA HUNCH 2026–27 student design project. Not a medical device.

GLADYS PORTER EARLY COLLEGE HIGH SCHOOL
NASA HUNCH 2026–27

LUNAR SPLINT DESIGN CHALLENGE

ORTHOPRINT

by Team Lunova



A custom-fit splint for a lateral ankle ligament tear, 3D-printed on demand on a lunar base.

SCHOOL #	_____
TEACHERS	Lupita C. Vasquez and Claudia Cortez
NASA PROJECT	Lunar Splint Design Challenge
TEAM PROJECT	Lunova · OrthoPrint

PRODUCT VALUE

PRINTED FOR ONE ASTRONAUT, WHEN THEY NEED IT

OrthoPrint is a custom splint for a lateral ankle ligament tear that a lunar base prints on demand from a scan of the astronaut's leg. Instead of launching a full set of fixed-size splints at about $1,000 per gram, the habitat prints one that fits.

TOP ATTRIBUTES

1. **Custom fit**
   Printed from a scan of the patient's limb.
2. **One-handed straps**
   Reusable zip-tie style straps the astronaut can tighten alone.
3. **Firm support**
   Solid shell, straps in the athletic-tape X pattern.
4. **Built for weeks of wear**
   Hygiene features for cleaning and skin checks.
5. **Thermal phase design**
   TPU flexible enough to wrap the leg, firm when needed.
6. **Made in the habitat**
   Printable on a habitat FDM or SLA printer.

THE PROBLEM

NO HOSPITAL ON THE MOON

Astronauts on a lunar base face a high risk of musculoskeletal injury, and a return to Earth takes days at minimum. Launching splints in every size wastes launch mass. A lunar base needs a custom splint it can print on demand from a patient scan.

$1M	1/6 g	−1–2%
launch cost per kilogram	lunar gravity	bone density per month

LUNAR-SPECIFIC INNOVATIONS

BUILT FOR THE MOON

INNOVATION 1

SINGLE-PIECE RATCHET CLOSURE

Zip-tie style straps with no loose parts, tightened or released with one hand.

INNOVATION 2

INTEGRATED HYGIENE

A breathable, sanitary design that can be cleaned and checked over weeks of wear.

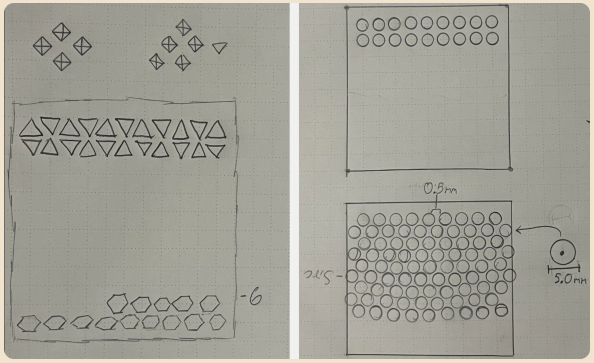
INNOVATION 3

THERMAL PHASE DESIGN

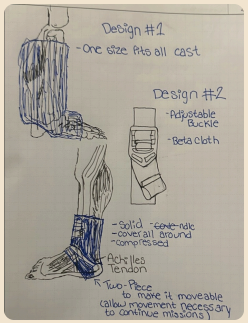
A TPU element flexible enough to wrap the leg and firm when needed.

DESIGN PROCESS

FROM SKETCH TO SPLINT



Hole patterns · Gertrudis Hernandez



Concepts 1 & 2 · Nancy De La Cruz

CONCEPT 1

One-size cast. A cast is not a splint, so we switched.

CONCEPT 2

Two pieces with an adjustable buckle.

TESTING

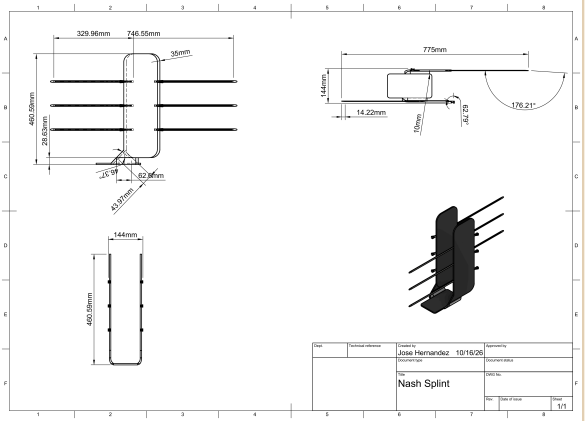
Holes made it too flexible, so the wall went solid.

NOW

Unlatches, fits on vertically, zip-tie straps in an X.

TECHNICAL SPECIFICATIONS

ORTHOPRINT	PROTOTYPE #1 · OCT 2026
INJURY	Lateral ankle ligament tear (3–26 weeks)
FIT	Custom, from a limb scan
CLOSURE	Reusable zip-tie style straps, X pattern
SHELL	Solid wall; small vents being re-tested
PRINTER	Habitat FDM or SLA
PROTOTYPE	PLA, ABS, TPU (PETG tested)
FLIGHT	CF-PEEK (recommended)
SIZE	460.59 mm tall; side plates 144 mm apart
MASS	To be measured from the final CAD
PRINT TIME	To be measured from the final CAD
SOFTWARE	None. OrthoPrint is a printed device



Detail drawing, dimensions in mm · Jose P. Hernandez, 10/16/2026

MATERIALS

PLA
Prototype 1

ABS
Prototype 2

TPU
Prototype 3 · flexes, firm when solid

PETG
Tested · broke when bent

CF-PEEK
The material we hope to use for flight